Lesson 4: Presenting Data and Recommendations

The last piece of technical reporting is the part that happens in the room. Everything up to this point — the cleaned workbook you built in Lesson 1, the one-page dashboard you built in Lesson 2, the structured deck you built in Lesson 3 — is preparation. In this lesson, you deliver. You frame the findings as a decision the audience needs to make, you annotate the charts so the takeaway is unmistakable, you adapt the same material for an executive or a technical audience without rewriting the data, and you handle the questions that come back at you with discipline rather than improvisation.

This lesson is also the capstone of the course. Everything you have built across the **OrbitCo** 30-day review comes together in a five-minute briefing: the Excel workbook is the evidence, the dashboard is the visual layer, the deck is the delivery vehicle, and you are the one whose name is on the recommendation. The capstone activity is integrated into this lesson, and its rubric is embedded at the end — it is the thing you are graded on, and it is also the thing that will be used to evaluate your reporting on the job. You will open the workbook you cleaned in Lesson 1 (the `Tickets`, `Uptime`, and `SLALookup` Tables plus the `outputs` Key Performance Indicator (KPI) summary), you will pull charts from the dashboard you built in Lesson 2, and you will embed them into a deck that follows the structure you learned in Lesson 3.

Two Registered Training Institute (RTI) standard topics carry through this lesson. "Use of productivity tools (Excel, PowerPoint) for technical reporting" is the umbrella — the briefing you deliver draws from both tools across all four lessons. "Data visualization" comes back explicitly here: choosing and annotating charts for the message, not just for the data, is the highest-value visualization skill and the one that separates a chart that exists from a chart that lands. On-the-Job Learning (OJL) competency 5d is tagged against this lesson because a well-constructed briefing is reusable operational documentation — the same deck can be refactored into onboarding material, a runbook appendix, or a status template for the next month's review.

Learning Objectives

By the end of this lesson, you will be able to:

1. Frame a set of findings as a data story tied to the audience's decision
2. Annotate charts and visuals to direct audience attention to the key takeaway
3. Adapt the same technical findings for different audiences (executive summary vs technical deep-dive)
4. Prepare for and handle Q&A, including defending claims with data and capturing follow-ups
5. Use PowerPoint presenter tools (speaker notes, rehearse with timing, basic accessibility settings)

Telling a Data Story

A data story is a set of findings organized around the decision the audience needs to make. That definition has two hinges, and both matter. **Organized** — because raw findings have no narrative; the author imposes one. **Around the decision** — because the structure is chosen to serve the audience's next action, not the author's analytical path.

A data dump is the opposite. A data dump walks through the analysis chronologically ("first we pulled the data, then we cleaned it, then we looked at severity, then we looked at Mean Time To Resolve (MTTR), then we looked at uptime"). It is accurate and it is useless. The audience has to assemble the story themselves, and they usually will not.

The Three-Part Arc

The simplest narrative structure for a data story is a three-part arc: **setup** (what is the situation), **conflict** (what is the problem), **resolution** (what should we do). Applied to the OrbitCo 30-day review:

- **Setup:** OrbitCo is committed to 99.9% uptime and a 60-minute MTTR on High-severity tickets. You have run that commitment for the last year and hit it every month.
- **Conflict:** Over the last 30 days, ticket volume has grown 22% and High-severity MTTR has crept from 48 minutes to 67 minutes. You are out of target on MTTR. Uptime is inside target but thin — any sustained incident could push you under.
- **Resolution:** Refactor the three runbooks causing the most repeat incidents. Re-evaluate the on-call staffing ask in 30 days with fresh data, after the runbook changes have had a chance to work.

The data supports every sentence of that arc. The arc is the organizing principle; the data is the evidence.

Leading with the Recommendation or with the Evidence

Two lead patterns for the same data story: **recommendation-first** or **evidence-first**. Which one you choose depends on the audience.

A senior decision-maker who trusts you and needs a fast answer wants the recommendation first: "I recommend we refactor the top three runbooks and re-evaluate the hiring ask in 30 days. Here is why." The evidence is the back-up for a decision the audience is already considering.

A technical peer who has to evaluate your analysis before they can endorse the conclusion wants the evidence first: "Here is the trend. Here is what we ruled out. Here is what remains. Therefore the recommendation is X." The evidence is the argument.

Neither lead is universally right. The skill is reading the audience and choosing.

When to Skip the Narrative

Not every operational presentation is a decision briefing. Three kinds of meetings warrant a more compressed format:

- **Urgent incident reviews** — the first 15 minutes after a Severity 1 (Sev-1) incident resolves. The format is facts, timeline, open questions. Narrative comes later in the post-mortem.
- **Structured status meetings** — recurring standups and weekly operations reviews. The format is the same every week: KPIs, exceptions, asks. No story is needed; the container is the story.
- **Deep technical working sessions** — a code review, an architecture deep-dive. The format is technical argument, not audience-facing narrative.

The story-arc advice in this lesson is for the briefing-style presentation. For the other formats, clarity and structure still matter; narrative does not.

Choosing and Annotating Charts for the Message

Every chart has two jobs. The first is to display the data accurately — that is Lesson 2's responsibility, and by now the OrbitCo dashboard charts do it. The second is to direct the audience's eye to the takeaway — that is this lesson's responsibility, and it is where most technical presenters under-invest.

The Single Takeaway Rule

Every chart in a briefing has one sentence that says what it shows. If the presenter cannot produce that sentence on demand, the chart is not ready. The sentence is not the chart title — it is the answer to "what is this chart telling us?" For the OrbitCo weekly trend chart, the takeaway sentence is "Ticket volume grew steadily over the 30-day window, with the largest jump in week 4." That sentence is the core of the chart's annotation.

Titles That Are Sentences

The chart title in a briefing deck is not a noun phrase ("Incident Trend" or "Weekly Tickets"). It is a sentence that names the finding. "Incidents dropped 40% after the runbook update." "High-severity MTTR has crept up over the last month." "Uptime is holding at 99.87%, thin of the 99.9% target."

A sentence title does two things a noun title cannot. It delivers the takeaway to readers who skim the slide without reading it in detail, and it forces the author to commit to a specific interpretation of the chart. Noun titles are vague because the author has not yet decided what the chart means; sentence titles require the decision.

Annotations That Direct Attention

Annotations are the on-chart markers that point the viewer at the important part. Three techniques, all of which PowerPoint and Excel support natively:

- **Callouts** — small text boxes with pointer lines, placed near the data point that matters. Examples: "Week 4: infra migration here" or "This row: Service Level Agreement (SLA) breach."
- **Highlighted bars or points** — color one bar or data point in the chart's accent color while leaving the rest in a neutral grey. The accent draws the eye.

- **Plain-language takeaway labels** — a short sentence placed inside or next to the chart area, stating the conclusion. Example: "MTTR jumped 40% in week 4 following the on-call rotation change."

A well-annotated chart can be understood in five seconds by a viewer who never hears the speaker. That is the bar.

What to Remove

Most default Excel charts contain elements that do not earn their place on a briefing slide. Removing them is as important as adding annotations:

- **Gridlines** that do not help the reader. A simple column chart usually reads better without horizontal gridlines.
- **Legend entries** for series the audience will not use. A chart with one visible series does not need a legend at all.
- **Redundant titles** — for example, a chart title that repeats the slide title.
- **Default color palettes** that include ten colors when only two series are shown.

The rule: strip the chart until every remaining element earns its place.

Applied to the OrbitCo Briefing

The weekly trend chart from the Lesson 2 dashboard lands on a briefing slide. On the way there, it gets a sentence title ("Incident volume trending up; MTTR slipping in parallel"), a callout on the week-4 data point ("Infra migration week"), and a plain-language takeaway label below the chart ("Volume growth is not the cause; the pattern shows runbook gaps"). Gridlines disappear. The legend shrinks to two entries. The chart now carries its meaning alone, even if the presenter is interrupted before they can narrate it.

Activity: The OrbitCo 5-Minute Capstone Briefing

In this individual exercise, you pull everything from Modules 1 and 2 together into a five-minute briefing for a simulated engineering-leadership audience. You open the workbook from Lesson 1, pull annotated charts from the Lesson 2 dashboard, build a five- to seven-slide deck using the Lesson 3 structure, and deliver it in Presenter View with a specific recommendation and a prepared set of anticipated questions. This is the capstone — the rubric at the end of this lesson is how the briefing is evaluated.

Adapting for Different Audiences

The OrbitCo 30-day review will get told to three audiences in the first month of its life: the support team itself (peers), the engineering leadership team (the decision-makers on staffing and runbook work), and the company's leadership (the executives who see the summary in a monthly all-hands). The underlying findings are the same for all three. The deck is not.

Executive Summary: 5 Minutes, 5 Slides Max

The executive version leads with the recommendation, supports it with two or three top-line facts, and ends with the ask. Five slides: title, recommendation, supporting evidence (top-line trend), cost and risk of the recommendation, ask and next steps.

A specific form: "We recommend refactoring the top three runbooks over the next two sprints and re-evaluating the on-call hiring ask in 30 days. Here is the 30-day incident data. Here is the cost (two sprints of engineering time). The ask is approval of the sprint allocation. Questions?"

No methodology slides. No slide explaining what MTTR means — if the audience does not know, the deck footnotes it. The focus is on the decision.

Technical Deep-Dive: 30 to 45 Minutes

The technical version walks the audience through the analysis in enough detail that a peer could replicate it. The same OrbitCo data, but now with the full methodology (how you cleaned the data, which tickets you excluded and why), the full severity breakdown (not just the headline), the alternative analyses considered (did you look at the data by team, or by resolution code?), and the explicit caveats (the 19 missing-severity tickets, the uptime window starting mid-week).

For a peer audience, all of this earns its place. The deep-dive deck will have 15 to 20 slides, detailed speaker notes, and a Q&A that runs long. That is fine — it is the right format for the audience.

Same Content, Different Depth and Lead — Never Rewrite the Data

A common mistake is to rewrite the data for different audiences — rounding for executives, keeping exact figures for technical peers. Do not do this. The data is the same everywhere. What changes is the depth of exposition (how much detail), the lead (recommendation-first or evidence-first), and the cuts (which facts make the edit and which go to the appendix).

If the executive and technical versions tell different stories about the same data, one of them is misleading. A better pattern: write one master deck, then cut the executive version from it. The cuts force you to answer "what is the minimum the audience needs to see?" which is a useful question at any audience level.

Handouts vs Slides

Slides support the room. Handouts support asynchronous review. The two have different design rules.

A slide with 11 bullet points is bad because the audience cannot read it while the speaker talks. A handout with 11 bullet points might be fine — the reader can work through it at their own pace. If your findings need detail that the slides cannot hold, produce a handout in addition to slides, not instead of them.

For the OrbitCo briefing, the deck is five slides; the handout is a two-page Portable Document Format (PDF) with the full KPI table, the severity-by-team pivot, the caveats list, and the link to the source Excel.

Speaker Notes

Speaker Notes (in the pane below each slide in PowerPoint's Normal view) are what you say about the slide — not what is on the slide. The audience does not see them. Their purpose is to preserve the narrative when someone else has to give your deck, to keep your key talking points handy during rehearsal, and to protect you against a blank-mind moment during delivery.

The rule: speaker notes are the sentences you will say; the slide is the visual that supports them. Write full sentences in speaker notes for anything you want to say the same way every time — the recommendation, the caveats, the key trend interpretation.

Preparing for and Handling Q&A

Q&A is where briefings are won or lost. A clean deck and a confident delivery can be undone by twelve minutes of unprepared Q&A; a modest deck can be saved by strong Q&A discipline.

Anticipating Questions

List the five hardest questions someone might ask before you deliver the briefing. For each, prepare the data that answers it. For the OrbitCo review, the five are predictable:

1. "Are we over or under our capacity target?" — prepare the ticket-per-engineer number.
2. "Is this a one-month anomaly or a trend?" — prepare the same metrics for the prior quarter.
3. "What is the cost of refactoring the runbooks vs hiring another engineer?" — prepare the engineering-time estimate and the on-call salary estimate side by side.
4. "Which teams are driving the volume growth?" — prepare the severity-by-team pivot.
5. "What happens if we do nothing?" — prepare a projection of the same trends forward 30 days.

Having the answer ready — and ideally the slide in your appendix — is the difference between a Q&A where you sound in command and one where you sound surprised.

"I Do Not Know" Is a Legitimate Answer

If a question comes in that you cannot answer, say so, specifically. "I do not know the team-level breakdown for resolution-code reasons. I will pull that tomorrow and send it." This is trust-building. What is not trust-building is inventing a number, hedging ambiguously, or answering a different question than the one asked.

Capturing Follow-Ups

Any question you agreed to follow up on needs to be captured somewhere the questioner can see later. A notes page on the final slide ("Follow-ups" with a bulleted list), a shared doc linked from the deck, or a tracker in the team's issue system. Pick a channel and commit to it. Follow-ups that fall into the void erode trust faster than any other Q&A failure.

Bridging Techniques

When a question is wandering, loaded, or off-topic, the bridging pattern is **acknowledge → answer the real question → connect back to the recommendation**. Example: "That is a fair concern about tool vendor choice. The bigger driver of our MTTR today is the runbook gap — tool changes do not help until the runbook work is done, which is why the recommendation puts runbooks first." The acknowledgment shows you heard the question; the reframe returns to your argument.

This is not deflection. It is staying on point. Deflection is ignoring the question; bridging is answering it briefly and then returning to what the briefing is actually about.

When a Question Exposes a Gap

Sometimes a question reveals that you missed something important. The right move is to say so, not to paper over it. "That is a real gap — I did not pull the data by region and that would change the recommendation if the growth is concentrated in one region. I will pull it and circle back by end of week." That answer costs you nothing. Paper-over answers cost you the audience's trust for the rest of the meeting.

Activity: Peer Feedback Round

In this small-group exercise, you pair up after the capstone briefings and exchange rubric-tied feedback on each other's delivery — two things that worked, one thing to change, and one follow-up question you would have asked if you had been in a real audience. The goal is to train both specificity-of-observation (as the giver) and hearing-feedback-without-defensiveness (as the receiver). Both skills matter on the job; this is the low-stakes place to practice them.

Presenter Tools and Basic Accessibility

PowerPoint has three features that improve delivery quality and four accessibility basics that improve who can consume the briefing. All seven take minutes to apply and compound over time.

Speaker Notes and Presenter View

Presenter View (Slide Show → Use Presenter View) shows your speaker notes, the current slide, and the next slide on your laptop while the audience sees the slideshow fullscreen on the projector. This is the delivery-time companion to speaker notes — the notes you wrote are now visible to you and invisible to the audience.

The trick: write notes for every slide in the form you will actually say. "Volume grew 22% over the prior period. The week-4 spike tracks the infra migration. MTTR slipped after the on-call rotation change in week 3." Present these as sentences, not as bullet fragments you would have to expand on the fly.

Rehearse with Timings

Slide Show → Rehearse Timings runs the slideshow and records how long you spend on each slide. At the end, you see the total time and the per-slide distribution. Two or three rehearsals through the deck give you a realistic estimate of your actual delivery length, catch the slides where you overran, and expose the slides you moved past too fast.

For a five-minute briefing, the per-slide budget is about 45 to 60 seconds. If Rehearse Timings shows one slide eating 90 seconds, the slide has too much content or you are over-explaining. Either cut the slide or cut the narration.

Export as PDF

File → Export → Create PDF produces a Portable Document Format (PDF) version of the deck for asynchronous distribution. The PDF is the handout. It preserves fonts and layout for every recipient regardless of PowerPoint version, and it opens on phones, tablets, and browsers without issue.

A PDF version should go out with every briefing — either beforehand (as pre-read) or afterward (as the record of what was shown). The deck stays in PowerPoint for editing; the PDF is the distribution artifact.

Accessibility Basics

Four accessibility practices that every technical deck should meet:

- **Alt text on images and charts.** Right-click → Edit Alt Text. A short sentence describing the chart's content ("Line chart showing weekly ticket volume rising from 9 to 17 over the 30-day window"). Screen readers use this; the sighted audience does not see it.
- **Logical reading order.** The tab order of slide elements should match the visual flow. **Home → Arrange → Selection Pane** lets you see and reorder. This matters for screen readers and for keyboard navigation.
- **Color contrast.** Body text against background should meet Web Content Accessibility Guidelines (WCAG) AA, roughly a 4.5:1 ratio. PowerPoint's **Review → Check Accessibility** flags insufficient contrast.
- **Font size floor.** 18pt body minimum, 28pt headline minimum, from Lesson 3. The accessibility tool will not flag small fonts specifically, but the audience notices.

Why Accessibility Matters Beyond Compliance

Accessibility is often framed as a compliance checkbox. The practical framing is better: accessibility is what makes your deck readable in a dim conference room, over a bad projector, on a phone during a long commute, by a reader in a hurry, or by a colleague who had cataract surgery last week. Every accessibility practice improves the deck for every reader, not just the ones with documented disabilities.

Capstone Rubric

The capstone is graded against four dimensions from the course outline. Every deliverable — the Excel workbook from Lesson 1, the PowerPoint briefing, and the live delivery — is evaluated against the rows that apply.

1. Data Handling — Accuracy, Appropriate Formulas, Clear Structure

- **Exceeds:** Tables used throughout; structured references everywhere; no hardcoded ranges; raw, processed, and outputs separated on different sheets; all anomalies (missing severity, duplicates, casing) handled with a documented choice; formulas chosen correctly (`COUNTIFS` where appropriate, `XLOOKUP` over `VLOOKUP`, `AVERAGEIF` used for severity-scoped MTTR).
- **Meets:** Data cleaned correctly; KPI formulas produce correct numbers; Tables used; one or two minor issues (for example, one hardcoded range) but nothing that breaks the analysis.
- **Below:** Hardcoded ranges that will break with new data; raw data edited in place (broken lineage); incorrect metric formulas; missing cleanup steps (duplicates still present, inconsistent severities).

2. Visualization — Chart-Type Fit, Readability, Honesty

- **Exceeds:** Chart types match the question every time (line for trend, column or bar for comparison, donut for composition with five or fewer slices, scatter for correlation); charts have sentence titles and at least one annotation; no 3D; no truncated axes without explicit marking; color used intentionally and accessibly; dashboard follows the tiles-trends-details layout; raw data is off the dashboard sheet.
- **Meets:** Chart types generally fit; titles present though some may still be noun phrases; no obviously misleading visuals; dashboard layout is clear.
- **Below:** Chart-type mismatches (pies with more than five slices, line chart for categorical data); 3D charts; truncated axes without explanation; raw data showing on the dashboard; default Excel titles and colors unchanged.

3. Presentation — Logical Structure, Audience-Appropriate Depth, Professional Design

- **Exceeds:** Deck follows problem → evidence → options → recommendation (or the inverted version for an executive audience); every slide has one main idea; Slide Master used consistently (one typeface family, one palette, aligned layouts); five to seven slides for the five-minute budget; WCAG AA contrast; alt text on charts and images.
- **Meets:** Clear structure and clear slides; some drift in typography or alignment but nothing that hurts readability; slide count reasonable for the time budget.
- **Below:** No clear structure; multiple main ideas per slide; walls of text; inconsistent fonts and colors; no alt text; low contrast; 10 or more slides for a five-minute talk.

4. Communication — Clarity of Recommendation, Supporting Evidence, Q&A Preparedness

- **Exceeds:** Recommendation stated in one clear sentence; every supporting claim tied to data in the deck or appendix; speaker notes used; briefing delivered within the five-minute budget; at least three of the pre-identified five audience questions answered confidently, with "I do not know, I will follow up" used for any that could not be answered.
- **Meets:** Recommendation present but possibly hedged or buried; evidence cited though links to the data may be loose; mostly on time; some Q&A preparation visible.
- **Below:** No explicit recommendation; evidence not tied to data; over time or under time by a significant margin; Q&A answered with deflection or invention.

A briefing is "complete and ready" when it scores Meets or better on all four dimensions. A briefing that scores Exceeds on all four is a briefing you can use as a template for future work.

Wrap-Up

Framing, annotation, audience adaptation, Q&A discipline, presenter tools. Those five skills are what move a deck from "I have slides" to "I delivered a briefing." The capstone activity is where you put them all together on the OrbitCo data — the Excel workbook from Lesson 1, the dashboard from Lesson 2, the structured deck from Lesson 3, delivered in a five-minute briefing to engineering leadership. Nothing else in this course depends on new tool skills; from here on, the work is practice and refinement.

Key Takeaways

- A data story is organized around the decision the audience needs to make, not around the order of the analysis. A data dump is the opposite and is almost always unhelpful, even when the underlying analysis is sound.
- Every chart in a briefing has one takeaway sentence. The chart title is a sentence, not a noun phrase; annotations direct the eye to the point; gridlines and default colors get stripped until every remaining element earns its place.
- Same data, different depth and lead. Cut an executive version from a master technical deck — never rewrite the numbers. If the two versions tell different stories, one of them is misleading.
- "I do not know, I will follow up by end of week" is a stronger Q&A answer than any guess or paper-over. Capture follow-ups in a channel the questioner can see later; the trust built by delivering on a follow-up compounds over months.
- Presenter View, Rehearse Timings, PDF export, and the four accessibility basics (alt text, reading order, contrast, font-size floor) take minutes to apply and compound across every deck you will ever deliver.

Reflection Questions

1. Think about the last briefing you gave or watched. Was it a data story or a data dump? What specifically made it one or the other, and what would you change in the first two minutes to move it toward a story?
2. Pick one of the five anticipated audience questions for the OrbitCo review. What data would you prepare to answer it, and where in your deck or appendix would

that data live?

3. The same OrbitCo findings will be presented to the support team, the engineering leadership team, and the executive leadership team in the next month. Which lead pattern (recommendation-first or evidence-first) would you choose for each audience, and what is the one-sentence justification?